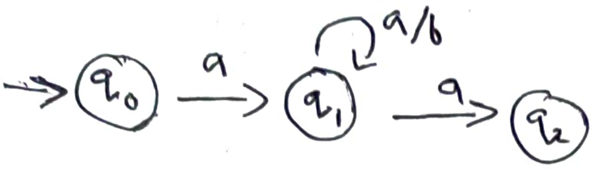
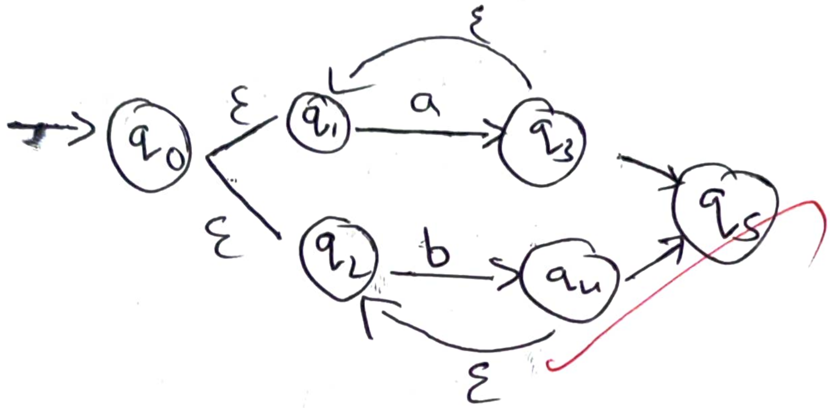
M. Vinay
192311146

56/100

2)

$a(a/b)^* a$





2

4)

$A \rightarrow A\alpha / \beta$

i) $E \rightarrow E+T/T$

ii) $T \rightarrow T * F/F$

iii) $F \rightarrow (E)/id$

i) $E \rightarrow TE'$

$E' \rightarrow +TE'$

ii) $T \rightarrow FT'$

$T \rightarrow * FT'$

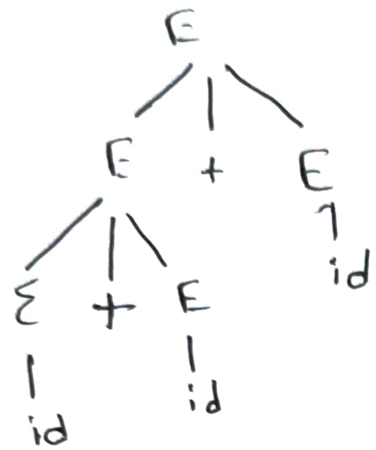
iii) $F \rightarrow (E)$

$F \rightarrow id$

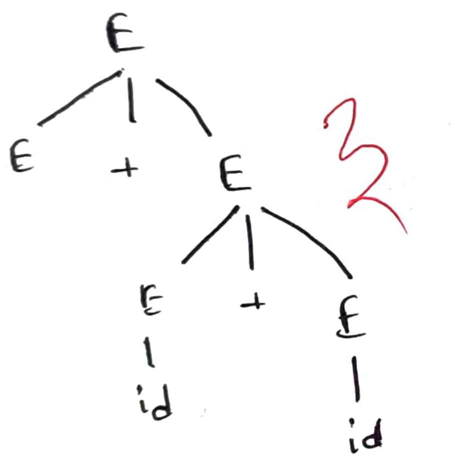
3

1) $E \to E+E / id$

$(id + id) + id$



$id + (id + id)$ — ①



3

$id + (id + id)$ — ②

① = ②

5) $S \to iEtS \mid iEtSeS \mid a$

$S \to iEtS\,S'$

$S' \to eS / \epsilon$

ii) left factoring

$S \to iEtS$

$S \to iEtSeS$

2

$S \to a$

$S \to iEtSS'$

$S' \to eS / \epsilon$

3) $E \to E + T / T$

$T \to T * F / F$

$F \to (E) / id$

$E \to E + T$

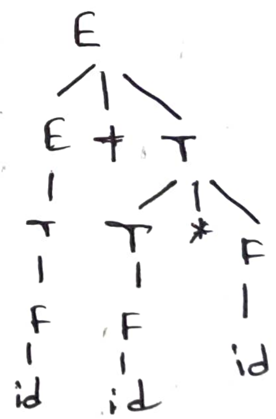
$\cancel{T + T}$

$F + T$

$id + T$

$id + T * F$

$id + id * F$



2

S → aBDh

B → cC

C → bC/ε

D → EF

E → g/ε

F → f/ε

2

First (S) = {a}

First (B) = {c}

First(C) = {b, ε}

First (D) = {g}

First (E) = {g, ε}

First (F) = {f, ε}

follow (S) = {$,}

follow (B) = {ε}

follow (C) = {ε}

follow (D) = {ε}

follow (E) = (ε)

follow (F) = (ε)

S → aBDh

B → cC

C → x

7) S → Sa | Sb | cd

A → Aα

S → Sa

S → Sb

S → ~~cd~~

it is immidiate left recursion

8) E → TE'

2 E' → TE'/ε

first (E') = {+, ε}

9) S → (S) S/ε

S → (S) S

((S) S) S

((ε)) ε

⇓

(())

Valid string

10) Shift:

We use shift to Eliminate terminals

11)

$S' \to S$

$S \to CC$

$C \to cC/d$

$S' \to S$

$S \to CC$

$C \to cC$

$C \to d$

I_0 : $S' \to .S$

$S \to .CC$

$C \to .cC$

$C \to .d$

goto (I_0, S)

I_1 : $S' \to S.$

goto (I_0, C)

I_2 : $S \to C.C$

$C \to .cC$

goto (I_0, c)

I_3 : $C \to c.C$

$C \to .d$

goto (I_0, d)

I_4 : $C \to d.$

goto (I_2, C)

I_5 : $S \to CC.$

goto (I_3, C)

$C \to cC.$

goto (I_3, d)

$C \to d.$

S → AB, A → a, B → b

First(S) → {A, B}

First (A) → {a}

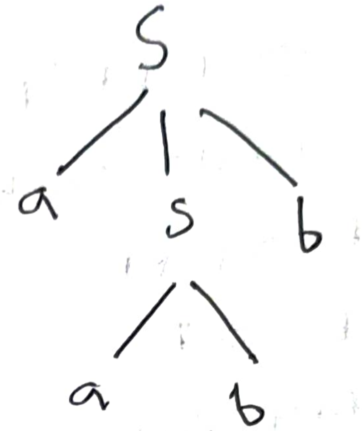
First (B) → b

S → asb / ab

aabb

S → asb

S → ab



3
2

* it accept aabb

13)

S → A

A → aB / Ad

B → b

C → g

First (S) = first (A) = {a, A}

first (B) = {b}

first (C) = {g}

follow (S) = {$}

follow (A) = {ε}

follow (B) = {ε}

follow (C) = {ε}

3
2

17)

Rules for left recursion :-

$E \rightarrow E+T/T$

separate each terminal

$E \rightarrow E+T$

$E \rightarrow T$

first perform left recursion with different Non-terminal

$E \Rightarrow T$

to Perform drag that E to other end

$E \rightarrow TE'$

Perform next step in that $E, E \rightarrow E'$

$E \rightarrow E+T$

$E' \rightarrow +TE'$

Rules for left factorial :-

3

$S \rightarrow iEts / iEtses / a$

take common Terminal a side

$\alpha a + \beta b$

$s \rightarrow iEts$

$(\alpha\beta)\ a+b$

After taking comomon their exist s'

$S \rightarrow iEts\ s'$

write the remaining

$S \rightarrow eS$

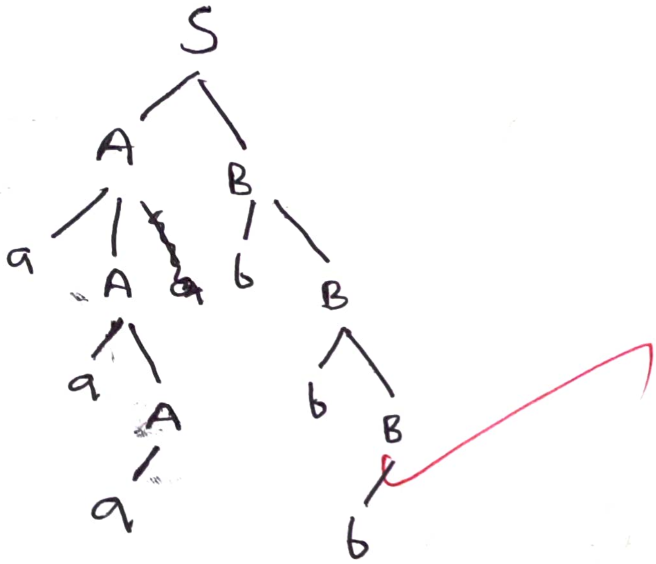
for next step there is no iEts so give ϵ

$S \rightarrow eS / \epsilon$

15) $S \rightarrow AB$

$A \rightarrow aA / a$

$B \rightarrow bB / b$



$= aaabbb$

2

18)

$E \rightarrow TE'$

$E' \rightarrow +TE' / \varepsilon$

$T \rightarrow FT'$

$T' \rightarrow *FT' / \varepsilon$

$F \rightarrow (E) / id$

First (E) = First (T) = First (F) = $\{ (, id \}$

First (E') = $\{ +, \varepsilon \}$

First (T') = $\{ *, \varepsilon \}$

Follow (E) = $\{ \$,) \}$

Follow (E') = $\{ \$,) \}$

follow (T) = $\{ +, \$,) \}$

follow (T') = $\{ +, \$,) \}$

follow (F) = $\{ +, \$,), * \}$

8

	+	*	(	)	id	$
E			$E \to TE'$		$E \to TE'$	~~$E \to TE'$~~
E'	$E' \to +TE'/\varepsilon$			$E' \to +TE'/\varepsilon$		$E' \to +TE'/\varepsilon$
T			$T \to FT'$		$T \to FT'$	
T'	$T' \to *FT'/\varepsilon$	$T' \to *FT'/\varepsilon$		$T' \to *FT'/\varepsilon$		$T' \to *FT'/\varepsilon$
F			$F \to (E)/id$		$F \to (E)/id$	$F \to (E)/id$

Stack	input	output
$E		
$E'T	id*id+id$	
$E'T'F	id*id+id$	$E \to TE'$
$E'T'id	id*id+id$	$T \to FT'$
$E'T'	id*id+id$	$F \to (E)/id$
$E'T'F*	*id+id$	$T' \to *FT'$
$E'T'id	*id+id$	$F \to (E)/id$
$E'T'F	id+id$	$T' \to FT'$
$E'T'F*	+id$	$E' \to +TE'$
$ E id	*id$	$F \to (E)/id$
$	id$	
	$	

19)

$E \to E + E$

$E \to E * E$

$E \to (E)$

$id1 + id2 * id3$

STack	input	~~output~~ Shift
$		
$id	id+id *id $	Shift
$ E	+ id * id $	reduce E->id
$ E+	+id *id$	Shift
$ E+id	id *id$	shift
$ E+E	*id $	reduce E->id
$ E	* id $	reduce E->E+E
$ E*	*id$	Shift
$ E*id	id $	
$ E*E	id $	reduce E->id
	id $	reduce E->E*E
$ $ E		accept

9